

AI Engineer Platform - Product Requirements Document (PRD) v1

Document Purpose

This document is the single source of truth for the AI Engineer Platform. Every engineering decision, feature, API, and architecture change should originate from this document before implementation.

Product Vision

Build a production-grade AI-powered engineering workspace that helps developers understand codebases, manage engineering knowledge, analyze repositories, generate documentation, and improve development productivity. This is not a generic chatbot; it is a developer productivity platform.

Problem Statement

Developers constantly switch between GitHub, documentation, ChatGPT, notes, PDFs, Stack Overflow, and IDEs. Context is fragmented and productivity suffers.

Target Users

Primary: Software Engineers, Full Stack Developers, AI Engineers. Secondary: Computer Science students and open-source contributors.

Project Goals

Production-quality architecture, clean code, reusable MERN modules, AI as an independent service, deployment-ready, Docker support, CI/CD, free-tier friendly infrastructure.

Non Goals (MVP)

No voice assistant, Slack integration, Jira integration, calendar, team collaboration, or complex multi-agent orchestration in version 1.

MVP Features

1. Authentication (JWT)
2. Dashboard
3. Workspace Management
4. AI Engineer Chat
5. Knowledge Base (PDF/Markdown/Text)
6. GitHub Repository Analysis
7. Documentation Generator
8. Bug Assistant
9. Notes & Saved Knowledge
10. User Profile & Settings
11. Usage Analytics

Architecture

Frontend: React + Vite + JavaScript + Tailwind

Backend: Node.js + Express

AI Service: Python + FastAPI

Database: MongoDB
Vector Store: To finalize
Deployment: Docker + GitHub Actions (later)

Engineering Principles

- Architecture first, code second.
- No temporary code.
- Modular structure.
- MERN remains the core application.
- AI runs as a separate service.
- Every feature must solve a real developer problem.
- Clean naming, validation, logging, error handling, documentation from day one.

Data Strategy

Store only metadata in MongoDB wherever possible.
Use configurable data-retention policies.
Use MongoDB TTL indexes for temporary data.
Keep AI cache and temporary processing data separate.

Development Workflow

PRD -> Architecture -> Database -> API -> UI -> Implementation -> Testing -> Documentation -> Deployment

Future Versions

Agentic AI, multi-agent workflows, MCP integrations, team collaboration, cloud deployment, advanced observability, plugins.

Roadmap

Phase 0: Product discovery & architecture
Phase 1: Project setup
Phase 2: Authentication
Phase 3: Dashboard
Phase 4: Knowledge Base
Phase 5: AI Chat
Phase 6: GitHub Analysis
Phase 7: Documentation Generator
Phase 8: Deployment
Phase 9: CI/CD

Decision Log

Maintain a DECISIONS.md file documenting every major architectural choice (database, authentication, AI service separation, Docker, TTL, etc.).